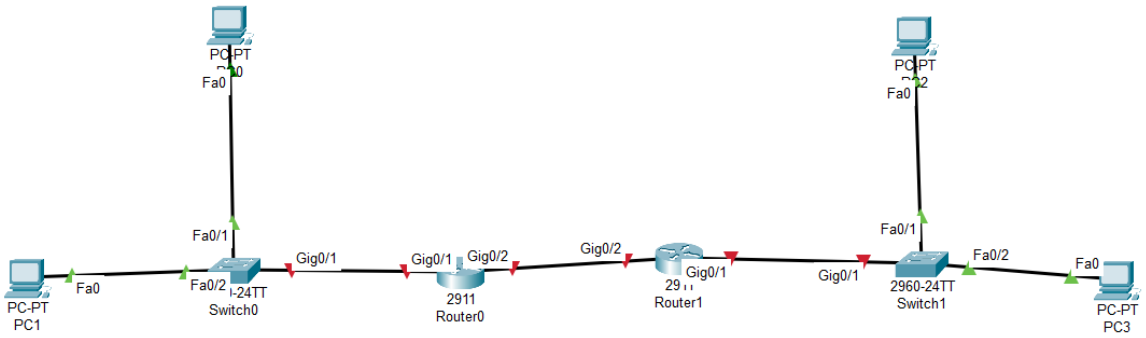
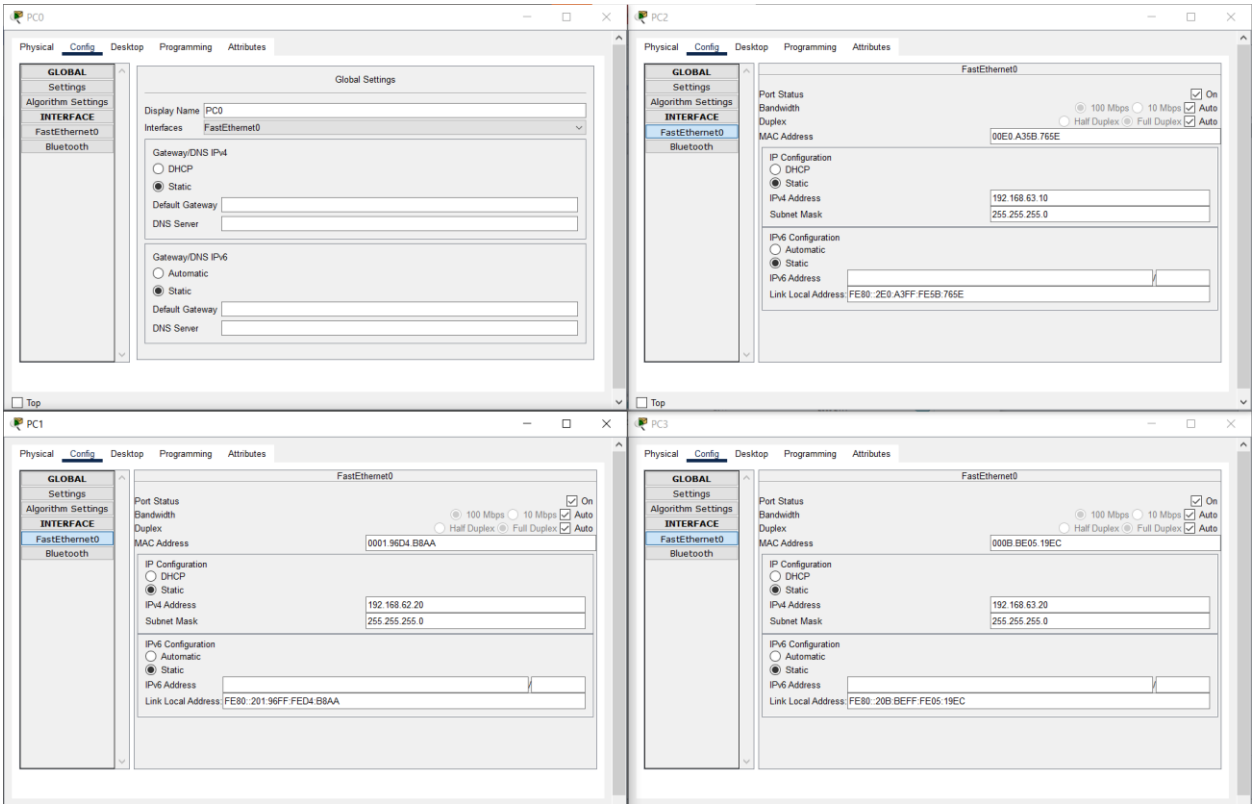
Практическая работа №3

Настройка топологии в CPT:



Назначим адреса для PC 0-3:



Базовая настройка маршрутизаторов:

Настройка R0:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#enable secret class
Router(config)#line console 0
Router(config-line)#password cisco
Router(config-line)#login
Router(config-line)#exit
Router(config)#interface gigab
Router(config)#interface gigabitEthernet 0/2
Router(config-if)#description Connected to Gi0/2 on R1
Router(config-if)#ip address 192.168.100.1 255.255.255.252
Router(config-if)#no sh
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

Router(config-if)#exit
Router(config)#hostname R0
R0(config)#interface gigabitEthernet 0/1
R0(config-if)#description Connected to SW1
R0(config-if)#ip address 192.168.62.1 255.255.255.0
R0(config-if)#no shutdown

R0(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R0(config-if)#exit
R0(config)#
```

Настройка R1:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#enable secret class
R1(config)#line console 0
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#exit
R1(config)#interfa
R1(config)#interface giga
R1(config)#interface gigabitEthernet 0/2
R1(config-if)#description Connected to Gi0/2 on R0
R1(config-if)#ip address 192.168.100.2 255.255.255.252
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

R1(config-if)#exit
R1(config)#interface gigabitEthernet 0/1
R1(config-if)#description Connected to SW2
R1(config-if)#ip address 192.168.63.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R1(config-if)#exit
R1(config)#
```

Эхо-тестирование:

Тестирующий Узел (IP)	Тестируемые узлы (IP-адреса)			
	PC1	PC3	R0	R1
PC0	+	-	+	-
PC2	-	+	-	+

Настройка маршрута по умолчанию:

R0:

```
R0>en
Password:
R0#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R0(config)#ip route 0.0.
R0(config)#ip route 0.0.0.0 0.0.0.0 192.168.100.2
R0(config)#
```

R1:

```
R1>en
Password:
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.100.1
```

Эхо-тестирование после настройки dafault routes:

Тестирующий Узел (IP)	PC1	PC3
PC0	+	+
PC2	+	+

Удаление default route:

На R0:

```
R0>en
Password:
R0#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R0(config)#no ip route 0.0.0.0 0.0.0.0 192.168.100.2
```

На R1:

```
R1>en
Password:
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#no ip route 0.0.0.0 0.0.0.0 192.168.100.1
```

Создадим vlan(10, 20) на Swintch0 и Switch1:

SW0:

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#exit
Switch(config)#interface range fast
Switch(config)#interface range fastEthernet 0/1 - 2
Switch(config-if-range)#switc
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#inter
Switch(config)#interface gig
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#swi
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Switch(config-if)#no shutdown
```

SW1:

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#exit
Switch(config)#int
Switch(config)#interface ra
Switch(config)#interface range fast
Switch(config)#interface range fastEthernet 0/1 - 2
Switch(config-if-range)#switchpo
Switch(config-if-range)#switchport mode acc
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#interfa
Switch(config)#interface gig
Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Switch(config-if)#no shutdown
```

Настройка Локальных интерфейсов Маршрутизаторов (Router-on-a-Stick):

R0:

```
R0>en
Password:
R0#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R0(config)#inte
R0(config)#interface gig
R0(config)#interface gigabitEthernet 0/1.10
R0(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.10, changed state to up

R0(config-subif)#enc
R0(config-subif)#encapsulation dot1q 10
R0(config-subif)#ip addr
R0(config-subif)#ip address 192.168.62.1 255.255.255.0
% 192.168.62.0 overlaps with GigabitEthernet0/1
R0(config-subif)#no shut
R0(config-subif)#no shutdown
R0(config-subif)#exit
```

R1:

```
R1(config-subif)#en
R1(config-subif)#encapsulation do
R1(config-subif)#encapsulation dot1q 20
R1(config-subif)#ip add
R1(config-subif)#ip address 192.168.63.1 255.255.255.0
% 192.168.63.0 overlaps with GigabitEthernet0/1
R1(config-subif)#no
R1(config-subif)#no s
R1(config-subif)#no sh
R1(config-subif)#no shutdown
R1(config-subif)#exit
```

Тестирующий Узел (IP)	Тестируемые узлы (IP-адреса)			
	PC1	PC3	R0	R1
PC0	+	-	-	-
PC2	-	+	-	-

Настройка динамической маршрутизации OSPF:

R0:

```
R0>en
Password:
R0#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R0(config)#router ospf 1
R0(config-router)#netw
R0(config-router)#network 192.168.62.0 0.0.0.255 area 0
R0(config-router)#network 192.168.100.0 0.0.0.3 area 0
R0(config-router)#exit
```

R1:

```
R1>en
Password:
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#192.168.63.0 0.0.0.255 area 0
                  ^
% Invalid input detected at '^' marker.

R1(config-router)#network 192.168.63.0 0.0.0.255 area 0
R1(config-router)#network 192.168.100.0 0.0.0.3 area 0
R1(config-router)#exit
```

Эхо тестирование с OSPF:

Тестирующий Узел (IP)	Тестируемые узлы (IP-адреса)			
	PC1	PC3	R0	R1
PC0	+	+	+	+
PC2	+	+	+	+